MTH636 – Game Theory

Lecture 28: Behavior Strategies, Mixed Strategies & SPE Existence

1 Outcome Equivalence of Behavior and Mixed Strategies

Theorem: Outcome Equivalence Result

In an extensive form game, every behavior strategy is outcome equivalent to a mixed strategy.

Proof. Consider player i and a behavior strategy b_i. Suppose player i has m many information sets $h_1, \ldots, h_m$.

Define the mixed strategy σ_i induced by b_i as follows:

$$\sigma_i(s_i) \;=\; \prod_{j=1}^{m} \beta_i\Big(s_i(h_j),\, h_j\Big) \qquad \forall\, s_i \in S_i.$$

Validity of the Induced Mixed Strategy

We first check that σ_i is indeed a valid (probability) mixed strategy.

Non-negativity: Since $\beta_i(s_i(h_j), h_j) \geq 0$ for all $j = 1, \ldots, m$, we have

$$\sigma_i(s_i) \;\geq\; 0.$$

Sums to one:

$$\begin{aligned}
\sum_{s_i \in S_i} \sigma_i(s_i) &= \sum_{s_i \in S_i} \prod_{j=1}^{m} \beta_i\Big(s_i(h_j),\, h_j\Big) \\
&= \prod_{j=1}^{m} \sum_{s_i \in S_i} \beta_i\Big(s_i(h_j),\, h_j\Big) \\
&= \prod_{j=1}^{m} (1) \qquad \text{(since } \beta_i \text{ is a behavior strategy)} \\
&= 1.
\end{aligned}$$

Hence σ_i is a valid mixed strategy.

Outcome Equivalence

Fix a leaf node z and suppose $h_1, \ldots, h_k$ are the information sets of player i on the path from the root to z. Also assume $a_1, \ldots, a_k$ are the actions of player i at those information sets.

Under b_i, player i's contribution to the probability of reaching z is:

$$\prod_{j=1}^{k} \beta_i(a_j,\, h_j).$$

Under σ_i, the contribution becomes:

$$\sum_{\substack{s_i \in S_i \\ s_i(h_j)=a_j \\ \forall\, j=1,\ldots,k}} \sigma_i(s_i).$$

Expanding using the definition of σ_i:

$$= \sum_{\substack{s_i \in S_i \\ s_i(h_j)=a_j \\ \forall\, j=1,\ldots,k}} \prod_{\ell=1}^{m} \beta_i\Big(s_i(h_\ell),\, h_\ell\Big)$$

$$= \sum_{\substack{s_i \in S_i \\ s_i(h_j)=a_j \\ \forall\, j=1,\ldots,k}} \prod_{\ell=1}^{k} \beta_i\Big(s_i(h_\ell),\, h_\ell\Big) \prod_{\ell=k+1}^{m} \beta_i\Big(s_i(h_\ell),\, h_\ell\Big)$$

$$= \sum_{\substack{s_i \in S_i \\ s_i(h_j)=a_j \\ \forall\, j=1,\ldots,k}} \prod_{\ell=1}^{k} \beta_i(a_\ell;\, h_\ell) \prod_{\ell=k+1}^{m} \beta_i\Big(s_i(h_\ell);\, h_\ell\Big)$$

$$= \prod_{\ell=1}^{k} \beta_i(a_\ell;\, h_\ell) \sum_{\substack{s_i \in S_i \\ s_i(h_j)=a_j \\ \forall\, j=1,\ldots,k}} \prod_{\ell=k+1}^{m} \beta_i\Big(s_i(h_\ell);\, h_\ell\Big)$$

$$= \prod_{\ell=1}^{k} \beta_i(a_\ell;\, h_\ell) \prod_{\ell=k+1}^{m} \sum_{s_i \in S_i} \beta_i\Big(s_i(h_\ell);\, h_\ell\Big)$$

$$= \prod_{\ell=1}^{k} \beta_i(a_\ell;\, h_\ell) \cdot \prod_{\ell=k+1}^{m} (1)$$

$$= \prod_{\ell=1}^{k} \beta_i(a_\ell;\, h_\ell)$$

$$= \prod_{j=1}^{k} \beta_i(a_j,\, h_j).$$

Therefore σ_i and β_i are outcome equivalent.

For the other part (every mixed strategy is outcome equivalent to a behavior strategy), a proof can be found in Solan's book. □

2 Existence of SPE in Extensive Form Games with Imperfect Information

Theorem: Existence of SPE

Every extensive form game with imperfect information has a Subgame Perfect Equilibrium (SPE).

Intuition

The method is **backward induction over subgames**. This will always result in an SPE.

2.1 Example: Sequential Game with Imperfect Information

Consider the following extensive form game. Player I moves first and chooses A or C. If A is chosen, Player II (who has an imperfect information set grouping two nodes) chooses ℓ or r (from each of her two decision nodes). If C is chosen, a chance node assigns probability $\frac{1}{4}$ to one branch and $\frac{3}{4}$ to another. The terminal payoffs are:

- Under branch A: if II plays $\ell\ell$: $(4, 1)$; if II plays ℓr: $(4, 1)$ (from the left node's ℓ) and $(7, 2)$ (from the right node's r); etc.
- Specifically the four outcomes under Player I choosing A and Player II's choices (ℓ, ℓ), (ℓ, r), (r, ℓ), (r, r) at her two-node information set yield terminal payoffs: $(4, 1)$, $(7, 2)$, $(6, 3)$, $(4, 10)$.
- Under branch C: with prob $\frac{1}{4}$, outcome $(8, 4)$; with prob $\frac{3}{4}$, outcome $(8, 8)$.

Expected payoff from branch C:

$$\tfrac{1}{4}(8, 4) + \tfrac{3}{4}(8, 8) = (8, \tfrac{1}{4} \cdot 4 + \tfrac{3}{4} \cdot 8) = (8, 7).$$

When solving by backward induction, Player I compares the payoff achievable under A (given Player II's best response in her information set) with the payoff $(8, 7)$ from C.

In the reduced tree (after computing the continuation value), Player I chooses A (giving access to the information-set subgame) or C (giving $(8, 7)$ directly).

2.2 Backward Induction and SPE: Battle of the Sexes Example

Consider the Man/Woman (M/W) sequential Battle of the Sexes game:

- Man (M) moves first: F (Football) or D (Dance).
- Woman (W) observes and moves: F or D at each of her nodes.

The terminal payoffs are (Man, Woman):

$$(2, 1), \quad (0, 0), \quad (0, 0), \quad (1, 2)$$

at the four leaves (corresponding to (F, F), (F, D), (D, F), (D, D)).

Nash equilibria of this game include:

- $\big(F,\,(F,F)\big)$ – this is an **NE** (Man plays F, Woman plays F regardless).
- $\big(F,\,(F,D)\big)$ – this is an **SPE** (Woman plays F after F and D after D, which is sequentially rational).

Note

$(F,(F,D))$ is the unique SPE found by backward induction. The NE $(F,(F,F))$ is not an SPE because after the Man plays D, the Woman playing F is not a best response in that subgame (she would get 0 instead of 2 by playing D).

3 Imperfect Information Game with Chance Nodes – Further SPE Analysis

Returning to the imperfect information example, consider a more elaborate tree where after Player I plays A or C:

- After A: Player II has a single information set (imperfect information – she cannot distinguish the two nodes). Her actions are ℓ or r at each node. Terminal payoffs: $(4,3),\ (7,2),\ (6,3),\ (4,10)$.
- After C: a chance node (prob $\frac{1}{4}$ left, prob $\frac{3}{4}$ right). Left subtree: Player II moves with actions L, R yielding (0_0), (8_4). Right subtree: Player I moves with actions a, b yielding (8_8), $\binom{0}{8}$.

Working by backward induction over the subgames yields an SPE in which $(Ca,\,rR)$ is an SPE.

Note

Perfect Bayesian Equilibrium is another refinement (not covered in this syllabus).

4 From Strategic Form to Extensive Form

- **Question:** Given a strategic form game G, can we obtain an equivalent extensive form game Γ?

Formally:

$$G \longrightarrow \Gamma \longrightarrow G \quad \text{(equivalence)}$$

Let

$$G = \Big\langle N,\,(S_i)_{i\in N},\,(u_i)_{i\in N}\Big\rangle$$

and assume that $S_i = \{s_i^1, \ldots, s_i^{m_i}\}$ for each i.